Topic Communities and Influential Keywords in Reddit Machine Learning Discussions

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Abstract

Online discussion platforms generate large volumes of text that can be transformed into networks for structural analysis. This study constructs a keyword co-occurrence network from recent Reddit posts collected directly from the MachineLearning subreddit. Rather than modeling users as nodes, the network represents keywords as nodes and connects two keywords whenever they appear together in the same Reddit post title. After data collection and preprocessing, two research questions are explored. First, the study identifies the most frequently discussed topics in recent Machine Learning posts through keyword extraction and frequency analysis. Second, the study investigates which keywords are the most structurally influential in the resulting network using community detection and centrality measures. The final network contains 35 keyword nodes, 39 co-occurrence edges, and 9 detected communities. Multiple metrics, including degree centrality, betweenness centrality, PageRank, and the clustering coefficient, are used to analyze the graph. The results show that Reddit Machine Learning discussions are concentrated around a small number of recurring concepts, especially open-source tools, large language models, and review-related terms. In addition, a few keywords occupy highly central structural positions and function as hubs or bridges between thematic regions. Overall, the study demonstrates that network analysis provides an interpretable way to study topic structure in online technical discussions.

Code Repository —
https://github.com/Harish-m-07/26SP_ILS_Z639_SMM

Introduction

Online discussion platforms produce large volumes of user-generated text that can be naturally modeled as networks. On Reddit, users create posts, respond to shared content, and discuss technical ideas in a threaded format. Even when the analysis is limited to post titles rather than full comments, these titles often contain concentrated signals about the main ideas being discussed. When repeated concepts are extracted and connected based on co-occurrence, the resulting graph can reveal how discussion themes are organized.

The MachineLearning subreddit is particularly suitable for this type of analysis because it contains a broad mix of discussions about models, benchmarks, tools, papers, conference reviewing, and implementation questions. These topics often appear in recurring combinations, which makes keyword co-occurrence a meaningful representation of the discussion space. Instead of studying direct user-to-user interaction, this project studies conceptual relationships between discussion topics.

The goal of this project is to construct and analyze a Reddit keyword network using data collected directly from Reddit. The dataset is created by retrieving recent posts from MachineLearning, extracting keywords from titles, and building a co-occurrence graph in which nodes represent keywords and weighted edges represent repeated co-occurrence across titles. Network analysis methods are then applied to understand both the content and structure of the discussion space.

Two research questions guide the analysis. The first asks which topics are discussed most frequently in recent Reddit Machine Learning posts. This question focuses on the descriptive distribution of content. The second asks which keywords are the most structurally influential in the discussion network. This question focuses on graph structure and influence patterns. By combining frequency analysis, community detection, centrality analysis, and network visualization, the study provides a structured view of current themes in Reddit Machine Learning discussions.

This project also satisfies an important course requirement: the dataset was not taken from Kaggle or from a published benchmark. Instead, the data was independently collected from Reddit and transformed into a network dataset through a reproducible workflow. The final result is a compact but meaningful graph that captures current topic structure within an online technical community.

Related Work

Network science provides a natural framework for studying online communication systems because interactions, entities, and concepts can be represented as graphs in which nodes correspond to meaningful units and edges represent relationships among them. This representation is useful because it shifts the focus from isolated observations to relational structure. Introductory work in network analysis emphasizes that graph-based approaches are especially valuable when the research goal is to identify structural proper-

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ties such as clustering, influential nodes, and global topology (Newman 2018; Barabási 2016).

One important approach in text-based network analysis is the use of keyword co-occurrence networks. In these networks, keywords are connected when they appear together in the same text unit, such as a title, abstract, article, or social media post. Co-occurrence graphs are useful because they convert unstructured text into a relational structure that can be analyzed with graph methods. In practical terms, they help identify topic clusters, central terms, and conceptual bridges between different thematic regions.

The first major methodological component of this project is community detection. Community detection attempts to partition a graph into groups whose internal connectivity is stronger than their external connectivity. Among the many available methods, modularity-based approaches are widely used because they provide an interpretable objective for discovering clusters in medium-sized graphs (Newman 2018). In a keyword network, communities often correspond to meaningful thematic groups. If a set of keywords repeatedly appears together, the corresponding nodes will tend to cluster within the same region of the graph.

The second major methodological component is centrality analysis. Centrality measures are used to identify structurally important nodes from different perspectives. Degree centrality captures how many direct neighbors a node has and is useful for identifying local hubs. Betweenness centrality highlights nodes that lie on shortest paths between other nodes and therefore connect otherwise distant parts of the graph (Freeman 1977). PageRank measures recursive influence by assigning higher scores to nodes connected to other important nodes (Page et al. 1999). These measures are especially relevant in a keyword network because topic importance can have multiple meanings. A keyword can be important because it is broadly connected, because it bridges multiple thematic regions, or because it is embedded near other influential keywords.

The third relevant methodological component is graph visualization. Force-directed layouts are widely used because they place connected nodes closer together and disconnected nodes farther apart, making hubs, clusters, and peripheral structures visually interpretable (Fruchterman and Reingold 1991). In addition, graph-level measures such as the clustering coefficient are useful for quantifying whether local groups of nodes tend to form tightly connected structures (Watts and Strogatz 1998). Together, these methods provide a strong foundation for studying Reddit keyword networks as both visual and quantitative objects.

Dataset and Data Processing

Data Collection

The dataset for this study was collected directly from Reddit using the public JSON endpoint for the MachineLearning subreddit. The workflow retrieved the newest 100 posts and extracted a structured table containing the post identifier, title, author, creation time, number of comments, and score. This direct collection process satisfies the requirement that the network dataset be independently constructed rather than taken from a preexisting source.

Although multiple metadata fields were collected, the primary textual source for analysis was the post title. Titles were chosen because they provide short summaries of the main discussion topic of each post. Unlike long comments or article bodies, titles are compact and focused, making them suitable for keyword extraction and co-occurrence analysis.

The resulting raw dataset consisted of 100 recent Reddit posts. This sample can be interpreted as a snapshot of current activity in the subreddit rather than a long-term historical archive. Even so, the collected posts provide enough variation to reveal recurring terms and keyword relationships.

Keyword Extraction and Transformation

After collection, the post titles were converted into a structured keyword dataset through a multi-step cleaning and transformation pipeline. The workflow first converted all titles to lowercase. Regular expressions were then used to extract alphabetic tokens while removing punctuation and mixed symbols. A manually defined stopword list removed very common words such as articles, pronouns, and generic function words that do not represent meaningful discussion themes. In addition, very short words were removed and duplicate words inside the same title were discarded so that a keyword could appear only once per post.

This process produced a cleaned keyword list for each post. The cleaned title representation was then used to construct two related datasets. First, a post-keyword edge list was created in which each row represented the fact that a particular keyword appeared in a particular post. This intermediate dataset documented the basic relationship between posts and extracted terms. In the notebook output, this stage produced 792 post-keyword edges.

Second, the post-level keyword lists were transformed into the final keyword co-occurrence network. For every title, all unique keyword pairs were generated. If two keywords appeared together in the same title, an undirected edge was created between them. Edge weights recorded how many titles contained that same keyword pair. In this way, the network captures repeated thematic association rather than a one-time accidental pairing.

Filtering and Final Network Construction

The raw co-occurrence graph contained many weak edges that reflected keyword pairs appearing together only once. Although such edges are not necessarily wrong, they produce a visually noisy graph and make the structural patterns harder to interpret. Therefore, the final network retained only keyword pairs with weight greater than or equal to 2. This filtering step removed one-time associations and kept only stronger recurring relationships.

After filtering, the final graph contained 35 keyword nodes and 39 edges. The network density was approximately 0.0655, the average degree was approximately 2.23, and the average clustering coefficient was approximately 0.4118. These values show that the graph is sparse overall but still contains meaningful local clustering. Many keywords appear in only one or two relationships, while a smaller number of terms connect multiple regions of the graph.

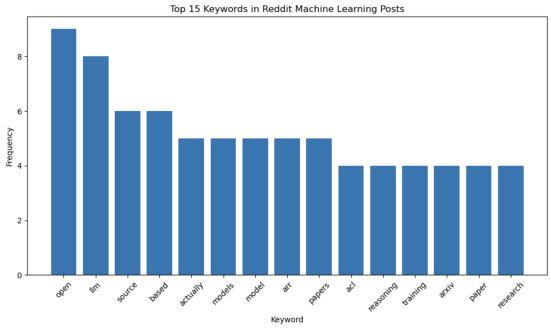

Figure 1: Top 15 keywords extracted from recent Machine-Learning post titles. The highest-frequency terms suggest strong emphasis on open-source topics, large language models, and review-related discussions.

The strongest keyword pairs in the final graph were *open–source* with weight 5, *acl–arr* with weight 4, and *llm–open* with weight 3. These early descriptive results already suggest the presence of interpretable thematic groups, including open-source tooling discussions, large language model discussions, and conference-review-related conversations.

Descriptive Analysis

Before addressing the research questions directly, it is useful to summarize the general structure of the dataset. The graph is not dense or uniformly connected. Instead, it contains a few highly connected terms and a larger number of low-degree or peripheral terms. This is also visible in the degree distribution shown later in the paper. Such a pattern is common in many real-world networks because a small number of nodes often attract a disproportionate share of the connections.

The descriptive analysis also shows that several extracted terms are closely tied to current Machine Learning discourse. Terms related to open-source releases and tooling appear prominently, as do large language model terms, model-development terms, and review-cycle terms such as *arr* and *acl*. These descriptive observations provide the foundation for the research questions that follow.

Research Question 1: Method and Findings

The first research question asks: *What are the most frequently discussed topics in Reddit Machine Learning posts?* This question focuses on identifying dominant themes in the collected titles rather than on graph structure itself.

To answer this question, all cleaned keywords extracted from the titles were aggregated into a single list and counted. The resulting frequency table was sorted in descending order to identify the most common terms. A bar chart of the top 15 keywords was then created for easier interpretation, as shown in Figure 1.

The results show that a relatively small set of terms dominates the current discussion space. The keyword *open* appears most frequently, followed by *llm*, *source*, and *based*. Other highly recurring terms include *actually*, *models*, *model*, *arr*, and *papers*. These findings suggest that the subreddit is strongly oriented toward open-source content, large language model developments, model-building concerns, and conference-related discourse.

Several of these keywords are especially interpretable when viewed in context. The pair *open* and *source* strongly suggests ongoing interest in public model releases, tools, and reproducible resources. The recurrence of *llm*, *model*, and *models* shows the continued prominence of language-model-related discussion. The visibility of *arr*, *acl*, and *papers* indicates that publication and reviewing topics also occupy a visible place within the subreddit.

An important point is that RQ1 provides a descriptive view of discussion prominence, but it does not by itself show which terms are structurally important in the network. A term can appear often but still be peripheral, while another can appear less often but connect multiple thematic areas. For this reason, the second research question examines structural influence more directly.

Still, the findings from RQ1 are useful because they establish the topical landscape of the dataset. Reddit Machine Learning discussions are not spread evenly across all possible concepts. Instead, they are concentrated around a limited set of themes that shape the broader discussion space. This concentration is one reason a keyword co-occurrence network is useful: recurring terms generate repeated pairings that produce interpretable network structure.

Research Question 2: Method and Findings

The second research question asks: *Which keywords are the most structurally influential in the Reddit Machine Learning discussion network?* Unlike the first question, which focuses on frequency, this question examines how keywords are positioned within the graph.

To address this question, two kinds of methods were applied. First, community detection was used to identify clusters of keywords that appear together more often than they connect to the rest of the graph. Second, multiple centrality measures were computed to identify nodes that function as hubs, bridges, or globally influential terms.

Community Detection

Community detection was performed using the Greedy Modularity Community Detection algorithm implemented in NetworkX. This method partitions the graph by maximizing modularity, which compares the density of edges inside candidate communities to the density expected under a random baseline (Newman 2018). Because the graph in this project represents keyword co-occurrence rather than directional flow, the undirected representation of the graph is appropriate for this analysis.

The algorithm identified 9 communities in the final network. Figure 2 shows the network colored by community membership. The visualization reveals that the graph contains several clear thematic regions rather than one undifferentiated cluster.

Several of the detected communities are interpretable. One prominent group includes *open*, *source*, *project*, and

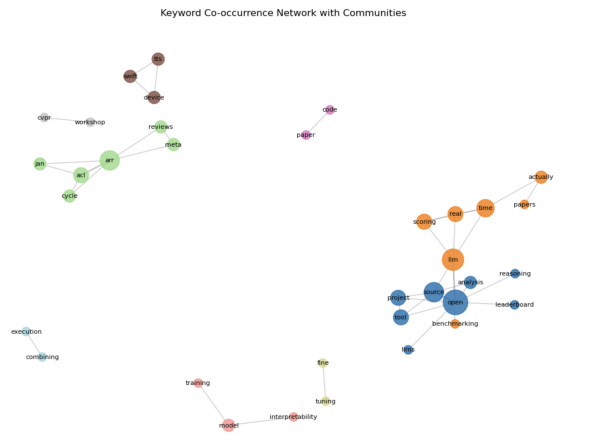

Figure 2: Keyword co-occurrence network colored by community. The detected communities indicate separate thematic regions such as open-source tooling, large language model discussion, review-cycle terms, and smaller peripheral topic groups.

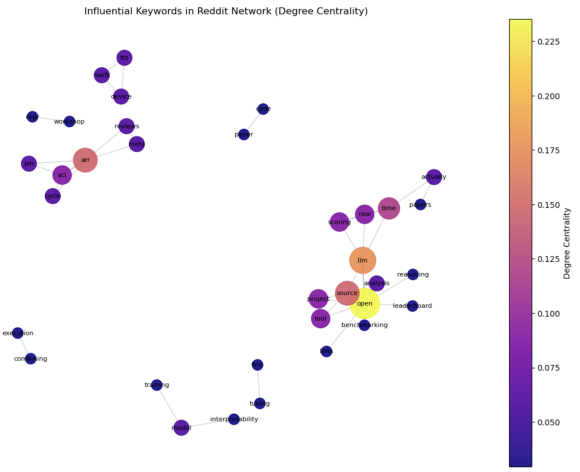

Figure 3: Keyword network scaled and colored by degree centrality. The terms *open*, *llm*, *source*, and *arr* are the strongest hub-like keywords in the graph.

tool, which together suggest open-source and tooling discussions. Another includes *llm*, *time*, *scoring*, *papers*, and *actually*, suggesting a language-model and evaluation-related region. A third includes *acl*, *arr*, *cycle*, *jan*, and *reviews*, clearly corresponding to conference-review and publication-cycle discussions. Smaller communities include terms such as *model*, *training*, and *interpretability*, as well as isolated pairs such as *code–paper* and *combining–execution*.

These results show that even a relatively small graph can contain coherent thematic substructure. The discussion space is therefore not random. Instead, recurring keyword pairings organize the graph into interpretable groups.

Centrality Analysis

After community detection, the analysis turned to node-level structural importance. Three node centrality measures were computed: degree centrality, betweenness centrality, and PageRank. In addition, the graph-level clustering coefficient was used to summarize local connectivity patterns.

Degree centrality identifies how many direct neighbors a node has relative to the size of the graph. In this network, the keyword *open* has the highest degree centrality (0.2353), followed by *llm* (0.1765), *source* (0.1471), and *arr* (0.1471). Figure 3 visualizes this result. These keywords act as local hubs because they connect directly to many other terms.

Betweenness centrality provides a different perspective. Rather than counting direct neighbors, it measures how frequently a node lies on shortest paths between other nodes. In this network, *llm* has the highest betweenness centrality (0.0945), followed by *open* (0.0847) and *time* (0.0428). Figure 4 shows that these nodes occupy positions that connect different parts of the graph. This means that *llm* functions as a particularly important bridge term. Even if another keyword has a higher local degree, *llm* is especially important for linking otherwise separate thematic regions.

PageRank captures global influence recursively by rewarding nodes that connect to other important nodes. In the notebook results, *open* has the highest PageRank (0.0816), followed by *arr* (0.0541), *llm* (0.0540), *source* (0.0503), and *model* (0.0417). Figure 5 shows this global influence structure. These results indicate that *open* is not only locally connected but also embedded in the most globally important part of the graph.

The graph-level clustering coefficient complements these node-level findings. The average clustering coefficient is 0.4118, which indicates a significant amount of local clustering among neighboring keywords. In other words, when a keyword is connected to multiple others, those neighbors are often somewhat likely to connect to each other as well. This is consistent with the presence of topic clusters in the graph.

Finally, the degree distribution shown in Figure 6 provides additional context. Most keywords have a low degree, while only a small number occupy more highly connected positions. This uneven distribution helps explain why a few terms stand out across multiple centrality measures.

Interpretation of Findings

The results of RQ2 show that structural influence in the keyword network is multidimensional. The keyword *open* is the strongest hub and the highest-ranked PageRank node, which indicates that it plays a central role in the global discussion structure. The keyword *llm* has the highest betweenness centrality, which means it plays a distinct bridging role between thematic regions. The keyword *arr* is especially important within the review-cycle cluster and still ranks highly in PageRank, showing that publication-related terms form a structurally meaningful region of the network.

These findings are important because they show that topic prominence and structural importance are related but not identical. Some terms matter because they occur often, while others matter because they connect different themes or oc-

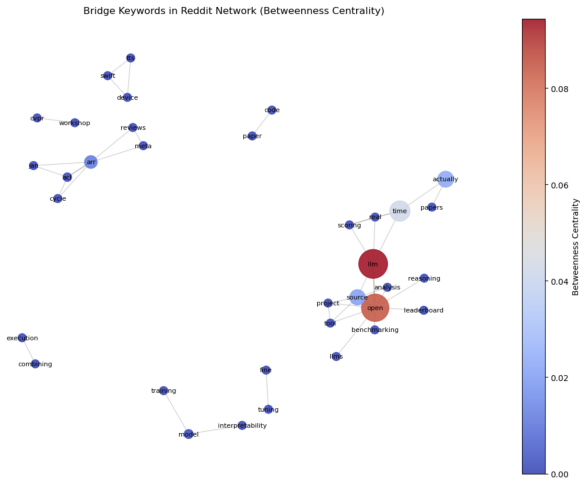

Figure 4: Keyword network scaled and colored by betweenness centrality. The keyword *llm* is the strongest bridge term, followed by *open* and *time*.

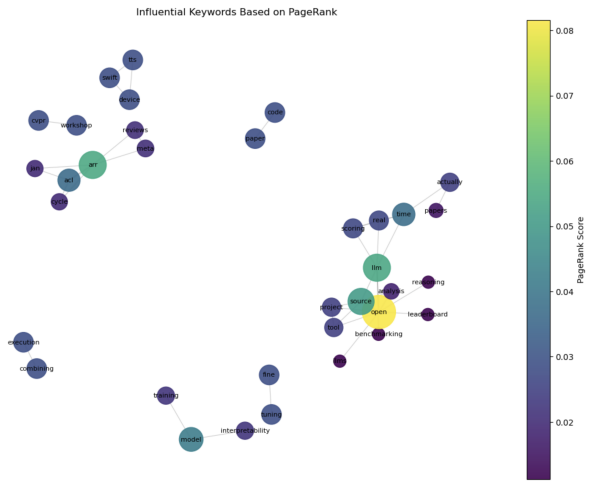

Figure 5: Keyword network scaled and colored by PageRank. The keyword *open* has the highest overall influence score, followed by *arr*, *llm*, and *source*.

cupy influential locations in the graph. This is why combining descriptive analysis with network analysis provides a stronger understanding of the dataset than either approach alone.

Discussion

Taken together, the two research questions provide a coherent view of current Reddit Machine Learning discussions. RQ1 shows which topics appear most often, while RQ2 shows how those topics are organized structurally. Frequency analysis reveals topical prominence, while network analysis reveals structural roles such as hubs, bridges, and central communities.

The community results suggest that the subreddit discussion space is organized around a few broad thematic regions. One region focuses on open-source projects and tools. Another centers on large language models and related evaluation language. Another clearly reflects conference-review and publication-cycle conversation. These clusters are intuitive and consistent with the types of topics that commonly dominate online Machine Learning communities.

The centrality results further show that not all keywords contribute to the graph in the same way. Some terms, such as *open*, function as hubs that connect directly to many other concepts. Some, such as *llm*, function as bridges between thematic regions. Others, such as *arr*, become prominent because they anchor a highly recognizable topic cluster. This distinction matters because it shows that discussion structure is shaped by multiple forms of importance rather than by a single ranking.

The descriptive statistics also provide useful context. The network is sparse overall, with density 0.0655, but it still contains a meaningful clustering coefficient and several coherent communities. This suggests that a relatively small dataset can still produce interpretable network structure when the representation is chosen carefully.

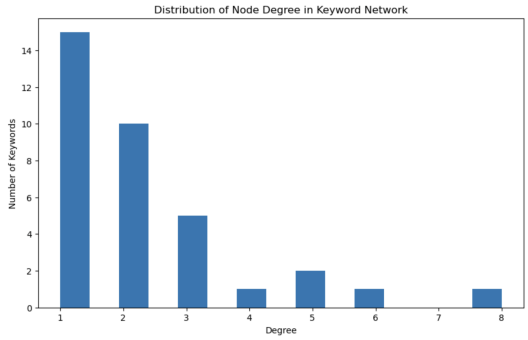

Figure 6: Distribution of node degree in the keyword network. Most nodes have low degree, while a small number of terms occupy more highly connected positions.

There are, however, important limitations. First, the analysis is based on only 100 recent posts, which means it reflects a short-term snapshot rather than a long-term history of subreddit activity. Second, the analysis uses post titles only and does not include the full post body or comment threads. Titles provide a focused signal, but they cannot represent all the semantic detail present in the full discussion. Third, keyword extraction was based on a manually defined stopword list rather than a more advanced linguistic pipeline. These limitations mean that the graph should be interpreted as a focused exploratory representation of current discussion themes.

Future work could extend the study in several directions. A larger dataset could be collected over a longer time period to examine whether communities remain stable or shift over time. The analysis could be expanded to include full comment content or multiple related subreddits. It would also be useful to compare the current keyword co-occurrence approach with phrase extraction, topic modeling, or temporal network analysis.

Conclusion

This project constructed and analyzed a Reddit keyword co-occurrence network using data collected directly from the MachineLearning subreddit. The workflow retrieved recent posts, extracted cleaned keywords from titles, built weighted keyword-pair relationships, and filtered the graph to keep stronger recurring associations. The final network contained 35 nodes, 39 edges, and 9 communities.

The first research question showed that recent discussions are concentrated around a small number of frequent topics, especially *open*, *llm*, *source*, and related terms. The second research question showed that structurally influential keywords include *open*, *llm*, *arr*, and *source*, each of which plays a different role in the graph as a hub, a bridge, or a globally central term.

Overall, the study demonstrates that network analysis is an effective framework for examining online technical discussion spaces. By transforming Reddit titles into a keyword co-occurrence graph, it becomes possible to identify both the dominant themes of discussion and the structural roles played by important concepts. Even with a relatively small sample, the resulting network provides a clear and interpretable snapshot of current Machine Learning discourse on Reddit.

References

Barabási, A.-L. 2016. *Network Science*. Cambridge University Press.

Freeman, L. C. 1977. A set of measures of centrality based on betweenness. *Sociometry*, 40(1): 35–41.

Fruchterman, T. M. J.; and Reingold, E. M. 1991. Graph drawing by force-directed placement. *Software: Practice and Experience*, 21(11): 1129–1164.

NetworkX Developers. 2026. NetworkX documentation. Available at https://networkx.org/documentation/stable/. Accessed 2026-03-15.

Newman, M. E. J. 2018. *Networks*. Oxford University Press.

Page, L.; Brin, S.; Motwani, R.; and Winograd, T. 1999. The PageRank citation ranking: Bringing order to the web. Stanford InfoLab Technical Report.

Reddit. 2026. Reddit API and developer documentation. Available at https://www.reddit.com/dev/api/. Accessed 2026-03-15.

Watts, D. J.; and Strogatz, S. H. 1998. Collective dynamics of small-world networks. *Nature*, 393: 440–442.

Code Availability

The source code and supplementary materials for this project are available at: https://github.com/Harish-m-07/26SP_ILS_Z639_SMM